GENERAL ANESTHETICS-II

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DIKIOHS

LEARNING OBJECTIVES GENERAL ANESTHETICS

1. Classify inhalational general anaesthetics.

2.Describe their mechanism of action.

Explain clinical uses, significance, and adverse effects.

3.Understand anaesthetic principles during maxillofacial surgeries, recognize effects on respiration and cardiovascular system, coordinate safely with anaesthesiologists in hospital dentistry.

4.Classify intravenous anaesthetics.

5.Describe mechanism of action and clinical uses.

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6.Identify adverse effects and contraindications

7.Understand IV sedation techniques, assist in dental procedures requiring deep sedation

NITROUS OXIDE

- INHALATION ANESTHETICS
- LAUGHING GAS
- WEAK GENERAL ANESTHETICS (LOW POTENCY ANESTHETIC)
- POTENT ANALGESIC
- NITROUS OXIDE COMBINE WITH OXYGEN
- ODOURLESS
- NON-IRRITATING
- COMMONLY USED IN DENTISTRY (50^% N_2O WITH OXYGEN)
- POOR BLOOD SOLUBILITY
- NOT DEPRESS RESPIRATION
- NOT PRODUCE MUSCLE RELAXATION
- NON TOXIC TO LIVER, KIDNEY, BRAIN
- SAFEST ANESTHETIC
- CAN CAUSE DIFFUSION HYPOXIA, PNEUMOTHORAX, PRESSURE IN SINUSES

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INTRAVENOUS ANESTHETICS

- Used For Rapid Induction
- Sole Agent For Short Procedures
- May Be Administered As Infusion To Provide Sustain Anesthesia During Longer Surgical Procedures
- Produces Loss Of Consciousness In One Arm- Brain Circulation (11 Sec)
- Low Doses Can Be Used For Sedation
- Induction Phase of Anesthesia Depends on Ionization, Lipid Solubility And Binding To Plasma Protein (arterial concentration of Unbound, lipophilic, unionized quickly transported to brain).
- Initially, the drug rapidly floods the brain and vessel-rich tissues.
- After a single bolus, metabolism and clearance play a very minimal role in early recovery.
- Following prolonged infusions or repeat doses, metabolism becomes crucial, and the drug accumulates in fat (adipose) tissue to form a reservoir, leading to delayed recovery.
- Redistribution of Drug From CNS Results In Early Recovery
- Repeated Infusion Causes Drug To Be Stored In Adipose Tissues Leads To Delayed Recovery
- Low CO Increases Drug Delivery To Brain

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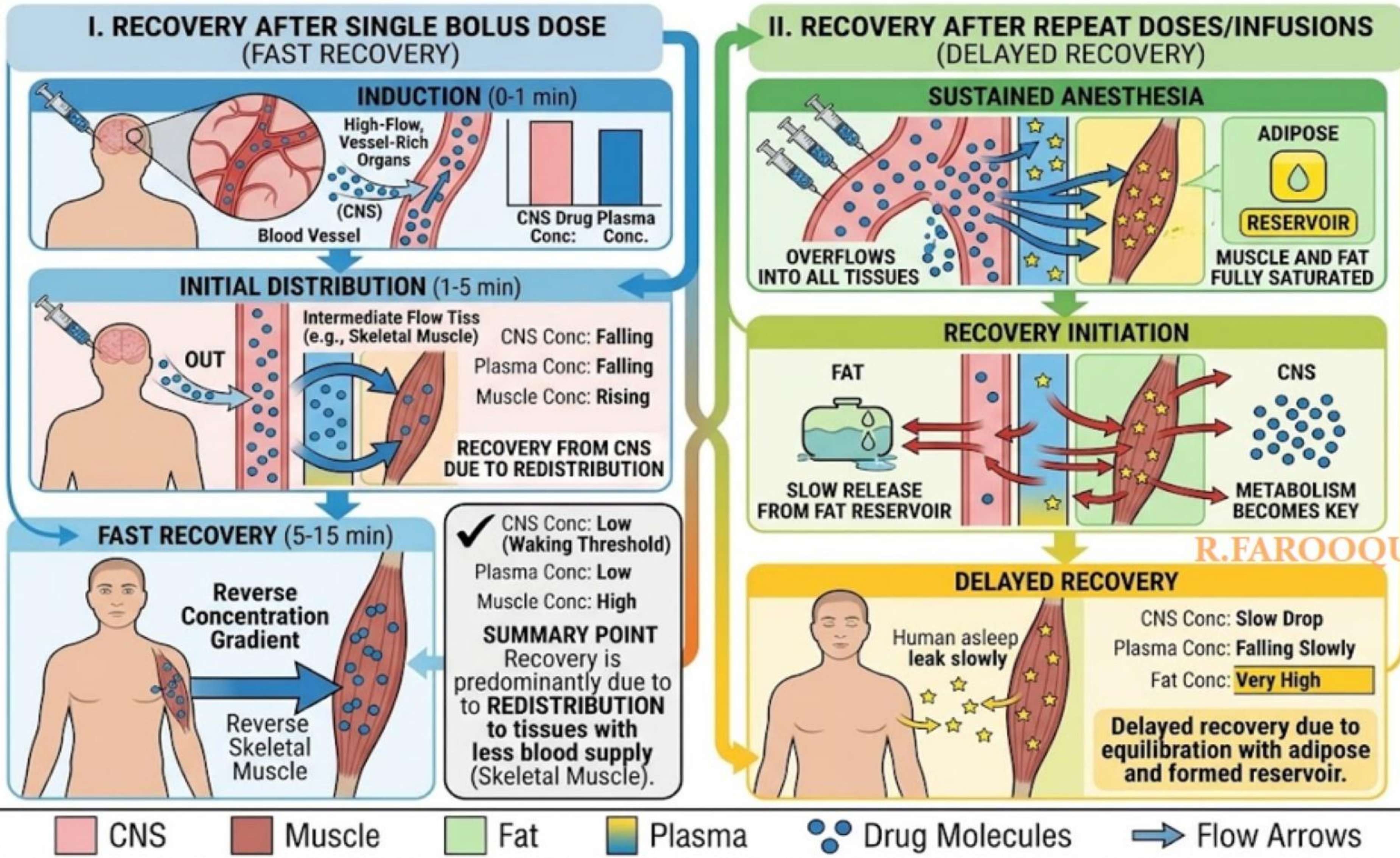
MECHANISMS OF RECOVERY FROM IV ANESTHESIA: SINGLE DOSE vs. REPEAT DOSES



SCENARIO

An 82-year-old patient with severe heart failure and significantly reduced cardiac output requires urgent endotracheal intubation. Which of the following is the most appropriate strategy for administering an intravenous induction agent in this patient?

A) Administer a standard induction dose rapidly to achieve quick unconsciousness.

B) Administer a higher induction dose because prolonged circulation time delays onset.

C) Administer a reduced induction dose and titrate it slowly due to altered blood flow distribution.

PROPOFOL

- IV Sedative /Hypnotic
- First Choice For Induction Anesthesia
- Widely Used, Has Replaced Thiopental
- Used For Both Induction And Maintenance
- Poorly Water Soluble, Supplied As An Emulsion containing Soybean Oil And Egg Phospholipid Given It Milk Like Appearance
- Onset Of Action 30-to-40 Seconds
- Plasma Level Declined By

1. Rapid Redistribution
2. Hepatic Metabolism
3. Renal Clearance
4. Initial Redistribution Half Life: 3-4 Min

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- No Effect Of Hepatic Or Renal Failure On Its Pharmacokinetics
- Thru Excitatory Phenomena It Depresses CNS By Muscle Twitching, Spontaneous Movement, yawning, hiccups, transient Pain at Inj Site.

PROPOFOL

- Decreases BP
- Reduces ICP (Due To Decrease Cerebral Blood Flow And Oxygen Consumption)
- No Analgesic Effect
- Has Antiemetic Effect (Therefore Low Incidence Of Nausea And Vomiting)
- In Lower Doses Used For Sedation
- Useful for Surgeries In Which Spinal Cord Function Is Monitored As It Has Less Of a Depressant Effect On CNS –Evoked Potentials.

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SCENARIO:

A 45-year-old patient undergoing a brief outpatient diagnostic procedure is given a single intravenous bolus of propofol for the induction of anesthesia. Despite the drug being metabolized slowly, the patient awakens fully alert within 10 minutes of administration.

A) Rapid hepatic metabolism and plasma clearance

B) Initial redistribution of the drug from the CNS to skeletal muscle

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C) Immediate saturation and storage of the drug within adipose tissue

SCENARIO

During a complex spinal surgery requiring continuous intraoperative monitoring of somatosensory evoked potentials (SEPs), the surgical team needs to maintain a stable depth of anesthesia without blunting neurophysiological signals. The anesthesiologist selects a propofol infusion over volatile inhalational agents for this procedure.

A) It accelerates peripheral nerve conduction velocity to bypass central pathways.

B) It exerts a minimal depressant effect on central nervous system (CNS) evoked potentials.

C) It selectively blocks peripheral pain fibers while leaving the dorsal columns completely unaffected.

KETAMINE

- Intravenous Anesthetics
- Short Acting
- Also provide analgesia
- Reduces OPIOID CONSUMPTION DURING SURGERY
- Lipophilic , Enters Brain Quickly
- NMDA receptor blocker
- Redistributed To Other Organs & tissues
- Produce Dissociative Anesthesia (Sedation, Amnesia, Immobility) Patient Is Unconscious But May Appear To Be Awake And Does Not Feel Pain.
- Increases Sympathetic Outflow
- Increases BP And CO
- Potent Bronchodilator
- Effective In Shock (hypovolemic and cardiogenic) Patients And Asthmatics
- Contraindicated In Hypertensive Patients Or Stroke Patients
- Increases Cerebral Blood Flow
- Induces Hallucination & dream like state (may be used illicitly)
- Used For Short Procedures

SCENARIO:

A 42-year-old opioid-tolerant patient undergoing major orthopedic surgery receives a low-dose ketamine infusion as part of a multimodal perioperative analgesic regimen. Postoperatively, the clinical team observes that the patient's cumulative morphine consumption is significantly reduced compared to their historical baseline.

A) Non-competitive antagonism of N-methyl-D-aspartate (NMDA) receptors

B) Irreversible blockade of voltage-gated sodium channels in peripheral nerves

C) Direct competitive antagonism at the central mu opioid receptors

ETOMIDATE

- Hypnotic Agent
- Used To Induce Anesthesia
- Lack Analgesic Effect
- Poor Water Solubility Therefore Formulated With Propylene Glycol Solution
- Rapid Induction
- Short Acting
- No Effects On Heart & Systemic Vascular Resistance
- Only Used For Patients With Cardiovascular Dysfunction I.E., CAD Or Critically Ill Patients
- It Inhibits 11-beta Hydroxylase Involved In Steroidogenesis
- Decreases Cortisol And Aldosterone Levels Which Persist For 8 Hours

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- Not Used For Longer Surgical Procedures
- It Causes Involuntary Muscle Movements, Nasua, Vomiting, Inj. Site Pain

SCENARIO:

A 68-year-old patient with severe coronary artery disease requires brief sedation for a cardioversion procedure, but long-term continuous infusion is avoided due to the risk of adrenal suppression. The anesthesiologist selects etomidate for its unique hemodynamic stability during induction.

A) It increases myocardial oxygen consumption and induces hepatic enzyme clearance.

B) It maintains cardiovascular stability and causes reversible inhibition of steroidogenesis.

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C) It produces profound peripheral vasodilation and permanently destroys adrenal cortical cells.

DEXMEDETOMIDATE

- Sedative, Anxiolytic, Analgesic, Sympatholytic Effects
- Alfa 2 Adrenoceptor Agonist In Certain Parts Of The Brain
- Used In ICU Settings And Surgery
- Provide Sedation Without Respiratory Depression
- It Reduces Volatile Anesthetic, Sedative And Analgesic Requirements Without Causing Significant Respiratory Depression
- It Blunts Emergence Delirium In The Pediatric Population
- Blunt Undesirable Cardiovascular Reflexes

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SCENARIO:

A 5-year-old child undergoes adenotonsillectomy under sevoflurane anesthesia. During recovery, the child becomes agitated, inconsolable, and disoriented despite adequate analgesia. The anesthesiologist administers dexmedetomidine to prevent this complication in future case.

A. Reduces postoperative nausea and vomiting
B. Blunts emergence delirium while providing sedation with minimal respiratory depression
C. Produces rapid recovery by increasing anesthetic elimination

SCENARIO:

A 60-year-old patient in the intensive care unit is receiving dexmedetomidine infusion for light sedation while remaining easily arousable and maintaining spontaneous ventilation.

A. Activation of GABA-A receptors in the cerebral cortex

C. Competitive antagonism of NMDA receptors

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B. Selective stimulation of central α_2-adrenergic receptors, reducing norepinephrine release

OPIOIDS

- Combined With Other Anesthetics
- Induce Analgesia
- I/V, I/T, E/D in to CSF
- Causes Hypotension
- Respiratory Depression
- Causes Muscle Rigidity
- Postanesthetic Nausea And Vomiting
- Not Good Amnesiacs

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- Commonly used opioids are Fentanyl, Sufentanil, Remifentanil (induce analgesia more rapidly than morphine)
- OPIOIDS effects can be blocked by Naloxone

SCENARIO:

A 58-year-old patient undergoing general anesthesia receives intravenous fentanyl for intraoperative analgesia. Shortly after surgery, the patient develops marked respiratory depression with a low respiratory rate and pinpoint pupils.

A. Flumazenil
B. Atropine
C. Naloxone

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BARBITURATES

- THIOPENTAL
- ULTRASHORTACTING
- HIGH LIPID SOLUBILTY
- POTENT ANESTHTICS
- WEAK ANALGESIC
- ONSET OF ACTION 1 MIN
- DIFFUSION OUT OF BRAIN IS RAPID
- REDISTRIBUTED TO OTHER TISSUES
- LONGER HALF LIFE BECAUSES OF SLOW METABOLISM

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- SEVER HYPOTENSION IN PATIENTS WITH SHOCK
- ADRS: APNEA, COUGHING, LARYNGEOSPASM,BRONCHOSPASM

BENZODIAZIPINES

MIDAZOLAM

- SEDATIVE, AMNESIC EFFECTS
- USED IN CONJUNCTION WITH OTHER ANESTHETICS
- FACILITATE GABA ACTION
- MINIMAL CVS DEPRESSANT EFFECT
- POTENTIAL RESPIRATORY DEPRESSANT EFFECTS
- INDUCE ANETROGRADE AMNESIA

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EFFECTS	PROPOFOL (SHORT ACTING)	KETAMINE	FENTANYL		MIDAZOLAM	THIOPENTAL
INDUCTION	30-40 SEC	1 MIN	-		5-10 MIN	1 MINUTE
RECOVERY	RAPID/REDISTRIBUTION OCCUR (2-4 MIN)	10-15 MIN	-		1 HOURS	CONSCIOUSNESS REGAINED 6-10 MIN
ANALGESIC	NO	YES	YES		POOR	POOR
CVS	HYPOTESION	HYPERTENSION (SYM STI)	HYPOTENSION		HYPOTENSION	HYPOTENSION
RESPIRATORY DEPRESSION	YES	NO	YES		YES WITH LARGE DOSES	YES WITH LARGE DOSES
AMNESIA	YES	YES	NO		YES	LITTLE AMNESIC
ANESTHETIC	SEDATIVE/HYPNOTIC	DISSOCIATIVE ANESTHESIA	NO LOSS OF CONSCIOUNSNESS, DROWSINESS		SEDATIVE/ HYPNOTIC	HYPNOTIC

EFFECTS	PROPOFOL	KETAMINE	FENTANYL		MEDAZOLAM	THIOPENTAL
VOMITING	ANTIEMETIC	EMESIS	EMESIS		EMESIS	EMESES
LIPID SOLUBILITY	HIGH	HIGH	HIGH		HIGH	HIGH
CONTRA-INDICATIONS	PANCREATITIS, SIEZURES ABNORMALLY LOW BP	HYPERTENSION, IHD, INC ICP	INCREASE ICP		HYPERSENSIT-IVITY, SHOCK	PORPHYRIA
INDICATIONS	1,DRUG OF CHOICE FOR INDUCTION 2. INDUCE SEDATION	ASTHMATICS, SHORT OPERATIONS, HEAD N NECK SURGEY,BURN DRESSING	SUPPLIMENTED WITH ANESTHETIC AGENTS		PREANESTHETIC, CONSCIOUS SEDATION,INDUCING, MAINTENANCE	COMMONLY USED INDUCING AGENT
ADRs	APNEA, DECREASE ICP, DYSTONIA,HYPOTENSION,SIEZURES	DELIRIUM, HALLUCINATION,INVOLUNTARY MOVEMENTS	DROWSINESS, MIOSIS, CONSTIPATION		HUICCUPS, DROWSINESS, VOMITING, HEADACHE	SHIVERING, DELIRIUM, RETLESSNESS

EFFECTS	PROPOFOL	KETAMINE	FENTANYL		MIDAZOLAM	THIOPENTAL
MOA	POTENTIATE GABA	BLOCKS NMDA	OPIOD AGONIST		POTENTIATE GABA	POTENTIATE GABA
DURATION OF ACTION	5-10 MINUTES	4 HOURS	50 MIN		6 HOURS	12 HOURS
PREGNANCY	SAFE	NOT CLASSIFIED	CATEGORY -C		BENEFITS OUTWEIGHS RISK	BENEFITS OUTWEIGHS RISK

THANK YOU

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